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Data Center Site Infrastructure Tier Standard:
Operational Sustainability

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Abstract

The Uptime Institute Tier Standard: Operational Sustainability is an objective methodology for data center owners to align the facility management program with the specific Tier of installed site infrastructure in order to achieve the organization's business objectives or mission imperatives. Tier Standard: Operational Sustainability establishes the behaviors and risks beyond the Tier Classification System (I, II, III, and IV) that impact long-term data center performance. Tier Standard: Operational Sustainability unifies site management behaviors with the Tier functionality of the site infrastructure.

Keywords

data center, infrastructure, tier, classification, tiers, tier level, topology, availability, reliability, redundant, concurrent maintenance, concurrently maintainable, fault tolerance, fault tolerant, operational sustainability, functionality, performance, metrics, tier standard, behaviors, risks, business objectives, mission imperatives, maintenance, failure response, critical load, capital investment, elements, management & operations, building characteristics, site location, design, construction, commissioning, transition-to-operations, basic capacity, redundant capacity, human error, staffing, organization, housekeeping, maintenance management system, service level agreements, life cycle, training, on-the-job training, planning, coordination, management, site policies, financial management, site infrastructure library, building features, design principles, operating conditions, natural disasters, man-made disasters, abnormal incident reports database, method of procedure, failure analysis, preventive maintenance, predictive maintenance, deferred maintenance, quality control, site configuration procedures, standard operating procedures, emergency operating procedures, purpose built, security, access, setback, set point, flood plain, seismic zone, risk evaluation

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Introduction

This introduction is not part of the Data Center Site Infrastructure Tier Standard: Operational Sustainability. It provides the reader with context for the application of the Standard.

Tier Standard: Topology (available separately) describes the functionality requirements of the site infrastructure to meet the specific business objective or mission imperative. Long-term availability of data center infrastructure is not guaranteed by Tier alone. Tier Standard: Operational Sustainability defines the behaviors and risks beyond Tier that impact the ability of a data center to meet its uptime objectives over the long term. The uptime of a data center is the resultant combination of both Tier of the site infrastructure and Operational Sustainability. This Standard is a tool to help owners maximize infrastructure investment. Additionally, this Standard facilitates comparison of data centers from an operational perspective.

Similar to Tier of installed infrastructure equipment, the rigor and sophistication of Operational Sustainability site management concepts and methodologies are established by the business requirements of the site. A Tier III performance requirement results in a more complex site infrastructure than a Tier I. Similarly, a Tier III data center requires more comprehensive behaviors and more rigor in mitigating risks than a Tier I. Therefore, Operational Sustainability behaviors and risk identification & mitigation are directly tied to the Tier Classification System.

The three elements of Operational Sustainability, in order of decreasing impact to operations, are Management & Operations, Building Characteristics, and Site Location. Each of these three elements has multiple categories and components with associated behaviors and risk. The Institute Abnormal Incident Reports (AIRs) database reveals that the leading cause of reported data center outages are directly attributable to shorfalls in management, staff activities, and operations procedures. Accordingly, Management & Operations is the most influential element to sustain operations.

Finally, Tier Standard: Operational Sustainability defines behaviors that result in more efficient operation of the data center, thereby offering opportunities to increase energy efficiency

Additional Factors and Exposures

Uptime Institute Tier Standard: Topology and Tier Standard: Operational Sustainability establish a consistent set of performance criteria that can be satisfied, and adjudicated, worldwide. For the data center design, implementation, and sustained operation to be successful, additional factors and exposures must also be considered by the owner and project team. Many of these will be dictated by the site location as well as local, national, or regional considerations and/or regulations. For example, building codes and Authorities Having Jurisdiction (AHJs); seismic, extreme weather (high winds, tornado); flooding; adjacent property uses; union or other organized labor force; and/or physical security (either as corporate policy or warranted by immediate surroundings).

Due to the many design and management options that may be dictated by the owner, regulated by local government, recommended by industry groups, or followed as a general practice, it is not feasible for Tier Standard: Topology and Tier Standard: Operational Sustainability to establish criteria for these additional factors and exposures worldwide. And, the Uptime Institute does not wish to displace or confuse the guidance of local experts, which are key for timely project delivery, regulatory compliance, and implementation of best practices.

For a successful project, Uptime Institute recommends that the project team create a comprehensive catalogue of project requirements, which incorporates Tier Standard: Topology, Tier Standard: Operational Sustainability, and carefully considered mitigation measures of these additional factors and exposures. This approach will ensure the project meets the compliance objectives of Uptime Institute's international standards, as well as local constraints and owner's business case.

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1. Overview

Scope

This Standard establishes the Uptime Institute Data Center Site Infrastructure Tier Standard: Operational Sustainability. This Standard establishes the behaviors and risks beyond Tier of installed infrastructure that impact the ability of a data center to meet its business objectives or mission imperatives over the long term.

This owner's Standard is complementary to the Tier Standard: Topology. Tier Standard: Topology establishes the performance requirements for configuration of the power and cooling equipment, including the capability for redundancy, planned maintenance work, or failure response without impacting the critical load. Tier Standard: Topology does not prescribe or constrain solutions. Rather the intent of Tier Standard: Topology is to provide the framework to align site infrastructure capital investment with the business objective(s) or mission imperative(s) that the data center supports.

Tier Standard: Operational Sustainability provides the site management behaviors and risks—contingent upon achievement of the site infrastructure. Consistent with the progressive nature of Tier Standard: Topology, Operational Sustainability behaviors increase in complexity and comprehensiveness as Tier increases.

The three elements of Operational Sustainability are Management & Operations, Building Characteristics, and Site Location. Each of these three elements has multiple categories and components with associated behaviors or risks. The specific behaviors, prioritized so that owners may address highest risks first, are presented in table format in this Standard.

The benefits of Operational Sustainability behaviors are fully realized when incorporated early into the project in the conceptual planning. Then, carried through design, construction, commissioning, and transition-to-operations—and ultimately addressed on a persistent basis during the operational life of the data center.

Purpose

Tier Standard: Operational Sustainability provides data center owners, operators, and managers with the prioritized behaviors and risks intrinsic to data center operations. Adherence to the recommended behaviors will assist in attaining the full performance potential of the installed infrastructure. This

Standard is a tool to help owners maximize infrastructure investment. Additionally, this Standard facilitates comparison of data centers from an operational perspective. Tier Standard: Operational Sustainability establishes a baseline of site management behaviors by Tier.

Tier Standard: Topology

Tier Standard: Topology establishes four distinctive definitions of data center site infrastructure using the Tier Classifications (I, II, III, and IV) and the performance confirmation tests for determining compliance to the definitions. The Tier Classification System describes the site-level infrastructure topology required to sustain data center operations, not the characteristics of individual systems or subsystems.

For informational purposes, the following is a brief summary of each Tier from the Tier Standard: Topology.

- Tier I – Basic Capacity: Site-wide shutdowns are required for maintenance or repair work. Capacity or distribution failures will impact the site.
- Tier II – Redundant Capacity components: Site-wide shutdowns for maintenance are still required. Capacity failures may impact the site. Distribution failures will impact the site.
- Tier III – Concurrently Maintainable: Each and every capacity component and distribution path in a site can be removed on a planned basis for maintenance or replacement without impacting operations. The site is still exposed to a equipment failure or operator error.
- Tier IV – Fault Tolerant: An individual equipment failure or distribution path interruption will not impact operations. A Fault Tolerant site is also Concurrently Maintainable.

Relationship Between Tiers and Operational Sustainability

Similar to Tier of installed infrastructure, the rigor and sophistication of Operational Sustainability site management concepts and methodologies are driven by the business requirements of the site. All three Operational Sustainability elements impact the performance potential of the Tier topology of the installed infrastructure, yet the Management & Operations Element has the largest impact on long-term availability. Staffing levels, the approach to maintenance, and the number and details of processes and procedures are illustrative categories of the Management & Operations Element that are directly related to each Tier level.

Exclusions From Operational Sustainability

Safety, environmental, and personnel management are not addressed in Tier Standard: Operational Sustainability. Failure to address any of these will add significant risk to data center operations. Yet, these items are excluded from the Standard because they are under the purview of a) management or internal corporate compliance audit groups and/or b) external enforcement and regulatory agencies.

Reference

Data Center Site Infrastructure Tier Standard: Topology (www.uptimeinstitute.com).

2. Elements of Operational Sustainability

Management & Operations

Analysis of the Institute AIRs database reveals that the majority of the reported data center outages are directly attributable to human error. Human error includes operator error—but more importantly, speaks to management decisions regarding staffing, maintenance, training, and overall rigor of the operation. The right number of qualified people is critical to meeting long-term performance objectives. Without the right number of qualified employees organized correctly, a data center does not have the resources to be successful.

After correct staffing, a comprehensive approach to maintaining a data center is an absolute requirement to achieve the uptime objective. An effective maintenance program encompasses increasingly rigorous preventive maintenance (PM), housekeeping policies, maintenance management system (MMS) to track work, and service level agreements (SLAs). As the performance objective increases, the requirements for documentation, complexity, and detail for each of these items increases.

In addition, a comprehensive training program ensures consistent operations and maintenance of a data center's infrastructure. All personnel must understand policies, procedures, and unique requirements of work in the data center to avoid unplanned outages and respond to anticipated events.

Building Characteristics

Building Characteristics include commissioning, building features, and infrastructure that can potentially affect attainment of availability objectives.

An extensive commissioning program is critical to a data center achieving a desired uptime objective. Comprehensive commissioning is the only way to ensure that the site infrastructure functions according to the design. It also provides the data center operators the opportunity to operate and test procedures without impacting the critical environment. Commissioning should exercise the equipment enough to identify potential manufacturing defects.

Building features can positively or negatively impact the availability objectives. Building features that support performance objectives include purpose-built data centers, adequate space for support and specialty spaces, and controlled access areas.

Additional infrastructure beyond just providing power and cooling is necessary to support the operation of a data center. Mechanical support systems such as chemical treatment and fuel scrubbing extend the life of a system and decrease the risk of failure. Having adequate space to safely conduct normal maintenance activities also reduce the risk of human error. The space, power, and cooling exhaust points must be aligned and monitored to avoid wasted capital expenditures.

Site Location

The highest level of functionality in a data center can be easily defeated by a local or regional disaster whether relating to natural occurrences or man-made factors. The site selection process for a new data center should evaluate the risks of these types of disasters. For new or existing data centers, these risks must be well documented, signed off by management, and with the proper level of mitigation in place.

Thus, both management expectations and the likely impact of the event on availability are accounted for. Depending on the performance objective, mitigation actions may be required.

3. Topology Enhancements

It is the data center owner's prerogative to enhance topology above that required for a specific Tier. Therefore, topology enhancements are not one of the Operational Sustainability Building Characteristics behaviors in this Standard. However, topology enhancements can significantly increase the performance potential by providing redundant capacity/distribution paths or Fault Tolerance above that required for a specific Tier. Having redundant components in a Tier IV System+System configuration reduces the risk of human error. Owners should consider practical topology enhancements for critical systems to support their Operational Sustainability program. The evaluation of topology enhancements should balance the synergies of having greater operational flexibility with greater operational complexity.

4. Behaviors & Risks

Table Organization

The tables in this Standard list and categorize the Operational Sustainability behaviors and risks. As the uptime objective (Tier) of a data center increases so does the number of Operational Sustainability behaviors required to achieve that objective. The behaviors applicable for each specific Tier are marked with a check.

The Site Location tables list the risks that should be evaluated. The tables also provide specific criteria to determine the magnitude of the risk. Develop mitigation plans for each risk identified.

Evaluating Effectiveness

The tables provide behaviors not requirements. There are many different ways to achieve these behaviors. It is important the behaviors both exist and are effective in order for the data center to achieve it uptime objective. There are three core principles to determine Operational Sustainability effectiveness: ***Proactive***, ***Practiced***, and ***Informed***. Evidence of all three principles must be present for a behavior to be considered effective.

Proactive

Is there a continuous improvement component present to ensure the processes and procedures are always being improved and kept updated? Behaviors are anticipated and appropriate processes and procedures are in place in advance. Evidence of a Proactive principle includes well-documented processes for all existing and anticipated activities with procedures in place for regular review and update.

Practiced

Are processes and procedures always followed? Having processes and procedures alone will not enhance Operational Sustainability unless all data center personnel consistently follow a disciplined approach. A task or procedure always accomplished the same way, no matter who is performing it, is evidence of this principle.

Informed

Is the knowledge to achieve a behavior held by the organization or an individual? Do all staff have knowledge of and access to all processes and procedures for any activity they might be required to perform? For example, does the maintenance technician required to perform a specific activity: 1) know there is a method of procedure (MOP) available for that activity, 2) where to find it, and 3) is granted access to it.

Prioritization

The prioritization of the Management & Operations and Building Characteristic behaviors are based on analysis of the AIRs database. Within each element, the categories and components are listed in the tables in order of decreasing importance. Site Location risks are of equal importance but specific criteria identifies the risk scale as higher or lower, based on the magnitude of potential impact. The level of mitigation in place will reduce potential impact to operations.

5. Summary

The behaviors established in Tier Standard: Operational Sustainability combined with the infrastructure requirements in the Tier Standard: Topology are essential for a site to achieve its uptime potential.

The installed infrastructure alone cannot ensure the long-term viability of the site unless Operational Sustainability behaviors are addressed. Site management teams that incorporate the principles of both Standards will have notably better results in realizing or exceeding the full uptime potential of the installed infrastructure.

6. Certification

The Uptime Institute reserves the exclusive right to rate and Certify data centers according to Tier Standard: Topology and Tier Standard: Operational Sustainability.

Please refer to uptimeinstitute.com.

Modifications

This Standard incorporates wording and organizational changes to clarify select behaviors. The Operational Sustainability Rating information is available at uptimeinstitute.com.

Staffing and Organization Category		Applicable for Tier			
Component	**Behavior**	**I**	**II**	**III**	**IV**
Staffing	Individual assigned full or part time to oversee critical facility operations	✓			
	Staff and/or vendors to support the business presence objective		✓		
	24 x 7 staff presence: minimum of 1 qualified FTE			✓	
	24 x 7 staff presence: 2 qualified FTEs of facility support per shift				✓
	Total FTE count numerically matches work load requirements		✓	✓	✓
	Escalation and call-out procedures are in place for assigned staff and specified vendor support for designated critical systems and equipment		✓	✓	✓
	Engineering trade (e.g., electrical, mechanical, controls, building management system [BMS], etc.) coverage split by shift based on operations and maintenance requirements			✓	✓
Qualifications	Appropriate staff trade licenses required by governmental regulation	✓	✓	✓	✓
	Experience and technical training required to properly maintain and operate the installed infrastructure			✓	✓
	Shift personnel qualified for specific shift operations individually and as a shift team			✓	✓
Organization	Organization chart showing reporting chain and all interfaces between the Facility, Engineering, Information Technology (IT), and Security groups	✓	✓	✓	✓
	Critical facility job descriptions—available and in use		✓	✓	✓
	Roles and responsibilities matrix covering all activities at the data center—available and in use			✓	✓
	Key individuals and alternates are designated			✓	✓
	Integrated approach to operational management, including all facets of the data center operation (Facilities, IT, and Security)			✓	✓

Table 1.1 Management & Operations–Staffing and Organization Category

Maintenance Category		Applicable for Tier			
Component	**Behavior**	**I**	**II**	**III**	**IV**
Preventative Maintenance Program	Effective preventive maintenance (PM) program including list of maintenance actions, due dates, and record of completion	✓	✓	✓	✓
	PM program encompasses original equipment manufacturer (OEM) maintenance recommendations		✓	✓	✓
	Detailed procedures for switching between redundant equipment—available and in use			✓	✓
	Fully scripted preventive maintenance activities (PMs) (e.g., through a method of procedure [MOP] process)			✓	✓
	Quality control process in place that validates a) the proper completion of and b) the quality of the PM			✓	✓
Housekeeping Policies	Computer room floor and underfloor free of dirt and debris	✓	✓	✓	✓
	Data center free of combustibles, cleaning equipment, shipping boxes, or personal conveniences items (e.g., coffee pots, microwaves)		✓	✓	✓
	Housekeeping policies available and enforced to ensure a contaminant free data center environment			✓	✓
Mainenance Management System	Effective maintenance management system ([MMS] paper or computerized) to track status of all maintenance activities—available and in use	✓	✓	✓	✓
	Maintains list of installed equipment (make, model, year of manufacture, year of installation, operating specifications, warranty information, etc.)		✓	✓	✓
	Work orders list special tools or parts required to complete PMs			✓	✓
	Maintains performance/trend data on equipment and history of maintenance activities			✓	✓
	Tracks calibration requirements			✓	✓
	Maintains list of critical spares and reorder points			✓	✓
Vendor Support	List of qualified vendors by system available for normal and emergency work	✓	✓	✓	✓
	Service level agreements (SLA) outlining scope of work, PM schedule, training requirements, and response times for all critical systems		✓	✓	✓
	Vendor call-in process and points-of-contact for pre-approved and qualified technician(s)			✓	✓
Deferred Maintenance Program	PM accomplishment rate greater than (>) 90%		✓		
	PM accomplishment rate of 100%			✓	✓
	Process for tracking deferred maintenance and executing it during a maintenance window	✓	✓		
Predictive Maintenance Program	Effective predictive maintenance program			✓	✓
Life-cycle Planning	Effective process for planning, scheduling, and funding the life-cycle replacement of major infrastructure components			✓	✓
Failure Analysis Prgram	Maintains list of all outages including dates, times, infrastructure equipment/ systems involved and specific computing outages, root-cause analysis, and lessons learned		✓	✓	✓
	Effective process to determine root cause, identify lessons learned, and implement corrective actions			✓	✓
	Trend analysis process				✓

Table 1.2 Management & Operations–Maintenance Category

Training Category		Applicable for Tier			
Component	**Behavior**	**I**	**II**	**III**	**IV**
Data Center Staff Training	On-the-job training (OJT) program for each new employee on a) the system(s) they will be responsible for operating and maintaining, and b) the rules of working in the data center	✓	✓	✓	✓
	Documented formal classroom, operational demonstrations, and/or shift drills covering the following: ▪ All policies, processes, and procedures for the operation and maintenance of data center systems ▪ Site Configuration Procedures (SCPs)—how the infrastructure is configured for normal operation ▪ Standard Operating Procedures (SOPs)—how the infrastructure configuration is changed during normal operations ▪ Emergency Operating Procedures (EOPs)—how the site is controlled and operated during abnormal circumstances or emergency situations ▪ MOPs ▪ MMS Procedures			✓	✓
	Training programs include training schedule, lesson plans, required reference materials, and records of attendance			✓	✓
	Formal qualification program for designated personnel performing data center operations			✓	✓
Vendor Training (Part-time Support)	List of training required before a vendor is allowed to work in the data center	✓	✓	✓	✓
	Briefing on data center processes and procedures with respect to the work to be performed		✓	✓	✓
	Formal training covering the appropriate subset of training received by the data center staff			✓	✓
	Training programs include training schedule, lesson plans, required reference materials, and records of attendance			✓	✓

Table 1.3 Management & Operations–Training Category

Planning, Coordination, and Management Category		Applicable for Tier			
Component	**Behavior**	**I**	**II**	**III**	**IV**
Site Policies	Formal documented policies and procedures for the following: ▪ Site staff performs all site infrastructure operations (e.g., configuration changes and operations under normal, emergency, or abnormal conditions) ▪ Site Configuration: Site infrastructure configuration for normal operations ▪ Standard Operations: Changes to normal operating configuration (e.g., shifting chillers) ▪ Emergency Operations: Control of the site during abnormal circumstances or event ▪ Change Management: a) review and approval of changes to the site baseline and b) evaluation of risk as related to planned changes ▪ Mitigation plans for site risks		✓	✓	✓
Financial Process	Process to ensure that operating and capital funding levels are consistently sufficient and available to support the business objective	✓	✓	✓	✓
	Operating and capital budgets managed separately from non-critical facilities and are not pooled with other buildings or groups of buildings			✓	✓
Reference Library	The following reference and record documents available for use (off site or on site): ▪ As-built drawings ▪ Operation and maintenance documentation ▪ Studies (e.g., soils, structural, electrical, mechanical, breaker, circuit, etc.) ▪ Commissioning reports ▪ Warranty documentation and pre-purchased maintenance agreements ▪ Written automation sequences of operation	✓	✓	✓	✓
	The previously listed reference and record documents available on site at all times			✓	✓
	Reference documents located in centralized location (library) available to site operational personnel			✓	✓
	Process ensuring master copies are maintained current with additional copies available to site operational personnel, vendors, designers, etc.			✓	✓
Capacity Management	Process for managing the installation and removal of IT equipment from the computer room	✓	✓	✓	✓
	Computer room floor plan—developed and regularly reviewed/updated		✓	✓	✓
	Process for forecasting future space, power, and cooling growth requirements on a periodic basis (e.g., 1/6/12/24/36 month)			✓	✓
	Tracking mechanism for current space, power, and cooling capacity and utilization reviewed periodically			✓	✓
	Effective process for a) computer room airflow management and b) electrical power monitoring, management, and analysis			✓	✓

Table 1.4 Management & Operations–Planning, Coordination, and Management Category

Operating Conditions Category — Applicable for Tier

Component	Behavior	I	II	III	IV
Load Management	Process to ensure the maximum loads are not exceeded and capacity is reserved for switching between components		✓	✓	✓
Operating Set Points	Consistent operating set points (e.g., temperature, pressure, volumetric flow, etc.) established based on **both** risk to continuous availability **and** cost of operation		✓	✓	✓
Rotating Redundant Equipment	Effective process for alternating the use of redundant infrastructure equipment as part of the site maintenance program		✓	✓	✓

Table 1.5 Management & Operations–Operating Conditions Category

Operating Conditions Category — Applicable for Tier

Component	Behavior	I	II	III	IV
Commissioning	Factory witness testing (FWT) of critical infrastructure equipment		✓	✓	✓
	Receipt, installation, and pre-functional testing of critical infrastructure components		✓	✓	✓
	Functional testing, critical infrastructure stand-alone testing, and pre-system startup configuration		✓	✓	✓
	System start, OEM test, and individual system test (IST)		✓	✓	✓
	Integrated systems operational test (ISOT)			✓	✓

Table 2.1 Building Characteristics–Pre-Operational Category

Building Features Category — Applicable for Tier

Component	Behavior	I	II	III	IV
Purpose Built	Purpose-built data center			✓	✓
	Single-purpose facility to support IT equipment operations			✓	✓
	Stand-alone building physically separated from other corporate facilities on the site			✓	✓
	Data center built to standards exceeding local building codes to ensure continued operations following a natural event			✓	✓
Support and Specialty Spaces	Adequate space separate from computer room for IT hardware receiving, storing, staging, building, and testing		✓	✓	✓
	Adequate space separate from computer room for the following functions: ▪ BMS/Building Automation System (BAS) control center ▪ Command Center/Disaster Recovery ▪ Parts and tool storage ▪ Engineering and facility shop activities ▪ Meeting and training purposes			✓	✓
Security and Access	Controlled access to all computer rooms and support spaces		✓	✓	✓
	Controlled building access			✓	✓
	Periodic review of access			✓	✓
	Controlled site access				✓
Setbacks	Adequate space around the data center to minimize impacts from adjacent facilities			✓	✓

Table 2.2 Building Characteristics–Building Features Category

Infrastructure Category		Applicable for Tier			
Component	**Behavior**	**I**	**II**	**III**	**IV**
Flexibility for Incremental Capacity Increases	Designed and constructed so that computer room space can be reconfigured with reasonable effort, and incremental increases in space, power, and cooling can be accomplished with minimal risk to the existing critical load			✓	✓
	Connection points for future/temporary extensions or capacity units			✓	✓
Infrastructure to Support Operations	Mechanical support systems available (e.g., chemical treatment, fuel scrubbing, etc.) to extend the life of or protect the infrastructure			✓	✓
	Mechanical systems installed to facilitate ease of operations			✓	✓
	Consistent labeling of infrastructure equipment and standardized sizes			✓	✓
	Electrical systems installed to facilitate ease of operations			✓	✓
Ease of Maintenance	Adequate space for the safe conduct of all normal maintenance activities on infrastructure equipment		✓	✓	✓
	Adequate space (sufficient swing radii, lifting points, and in/out pathways) for the safe conduct of rapid removal and replacement on infrastructure equipment			✓	✓
	Equipment access provided to facilitate delivery and installation of motors or other large components			✓	✓
Space, Power, and Cooling Exhaust Points	Data center design coordinated space, power, and cooling capacity exhaust points			✓	✓

Table 2.3 Building Characteristics–Infrastructure Category

Natural Disaster Category	Scale of Risk[1]	
Component	**Higher**	**Lower**
Flooding (river, lake, reservoir, canal, pond, etc.) and tsunami[2]	<100-year flood plain	>100-year flood plain
Hurricanes, tornadoes, and typhoons	High	Medium
Seismic activity[3]	>0.8 m/s^2	<0.8 m/s^2
Active volcanoes	High	Medium

Table 3.1 Site Location–Natural Disaster Category

Man-made Disaster Risk Category	Scale of Risk[1]	
Component	**Higher**	**Lower**
Aiport/Military airfield	<3 miles from any active runway; inside a 1x5 mile runway extension	>3 miles from any active runway; outside a 1x5 mile runway extension
Adjacent properties exposures	Chemical plant, fireworks, factory, etc.	Office building, undeveloped land, etc.
Transport corridors	<1 mile	>1 mile

Table 3.2 Site Location–Man-made Disaster Risk Category

[1]The level of mitigation in place will reduce potential impact to operations.

[2]Risk evaluation from the regional or local flood plain map or international equivalent.

[3]Peak ground acceleration (meters per second squared [m/s^2]) that can be expected during the next 50 years with 10% probability.